RMO 2025a

KV or JNV

Exercise 1 (RMO 2025a P1,). Solve the following system of equations in nonnegative integers $a_1, a_2, \ldots, a_8$ where $a_i \neq 1$ for $i = 1, \ldots, 8$:

$$\begin{aligned} a_1 a_2 &= a_3 + a_4, \\ a_3 a_4 &= a_5 + a_6, \\ a_5 a_6 &= a_7 + a_8, \\ a_7 a_8 &= a_1 + a_2. \end{aligned}$$

> **Walkthrough** —
>
> **(a)**

Solution 1.

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Exercise 2 (RMO 2025a P2,). Let a, b, c be positive real numbers satisfying $abc = 1$. Prove that

$$\frac{2a^2}{a^3+1} + \frac{2b^2}{b^3+1} + \frac{2c^2}{c^3+1} \leq a^2 + b^2 + c^2.$$

> **Walkthrough** —
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> **(a)**

Solution 2.

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Exercise 3 (RMO 2025a P4,). A frog is initially at $(0, 0)$ and it reaches $(n, 2)$, $n \geq 1$, using the following moves in any order several times:

1. $R = (1, 0)$, that is, if the frog is at (a, b) it goes to $(a + 1, b)$;

2. $U = (0, 1)$, that is, if the frog is at (a, b) it goes to $(a, b + 1)$;

3. $D = (1, 1)$, that is, if the frog is at (a, b) it goes to $(a + 1, b + 1)$.

In how many ways can the frog go from $(0, 0)$ to $(n, 2)$, $n \geq 1$, using the above steps subject to the condition that steps of the type UU, DD are forbidden? (For example, for $n = 3$, $RDUR, DRD$ are admissible paths, while $DDR, RUURR$ are not.)

> **Walkthrough** —
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> **(a)**

Solution 3.

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Exercise 4 (RMO 2025a P6,). Define the sequence $\langle a_0, a_1, a_2, \dots \rangle$ as follows: $a_0 = 49$ and $a_n = 10^{2^n} a_{n-1} - 1$ for $n \geq 1$. Show that $s(a_n^2) = n^2 + n + 7$ for all $n \geq 0$, where $s(m)$ denotes the sum of digits in base 10 representation of a nonnegative integer m.

> **Walkthrough** —
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> **(a)**

Solution 4.

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